

Predicting NBA Playoff Appearances Using Team Statistics

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1 Introduction

Every NBA season, 30 teams play through a full regular season, but only 16 make the playoffs. Making the playoffs is the first real step toward competing for a championship, and it also affects things like fan interest, media attention, and how teams make decisions going forward. For anyone who follows basketball, it naturally raises the question of what actually separates playoff teams from the rest of the league.

This project is meaningful to us because basketball is a sport we genuinely follow and have grown up watching. Several members of our group pay close attention to how NBA teams are built and evaluated, and we have noticed how the conversation has shifted from simple stats like points per game toward efficiency metrics like net rating and margin of victory. We wanted to go beyond just watching the games and actually test, using real data and the machine learning tools we have learned this semester, which statistics most reliably predict which teams earn a playoff spot. That question sits at the intersection of a sport we care about and methods that are directly relevant to this course.

Our goal is to use team statistics from the past six seasons to identify which metrics are most important for predicting playoff qualification, and to compare how well different machine learning models perform on this task.

Research Question: Which team statistics are most important in predicting whether an NBA team makes the playoffs?

2 Data Provenance

2.1 Who

The data comes from Basketball Reference (basketball-reference.com), a public sports statistics website operated by Sports Reference LLC. Basketball Reference aggregates official NBA statistics sourced directly from the league's data feed and is one of the most widely used public databases in basketball research and journalism (Sports Reference, 2000).

2.2 What

The dataset contains season-level team statistics for all 30 NBA franchises across six seasons: 2019–20 through 2024–25. Each row represents one team’s performance across a full regular season. After cleaning, the final modeling dataset contains **180 team-season observations** and includes the following variables:

- **Wins (W) and Losses (L):** The team’s regular season record
- **Margin of Victory (MOV):** Average point differential per game
- **Offensive Rating (ORtg):** Points scored per 100 possessions
- **Defensive Rating (DRtg):** Points allowed per 100 possessions (lower is better)
- **Net Rating (NRTg):** Offensive Rating minus Defensive Rating — the primary per-possession efficiency summary
- **Playoff:** Response variable — whether the team qualified for the postseason (Yes or No)

Two additional variables present in the raw files — Pace and Free Throw Rate — were excluded because they are missing for the 2019–20 and 2020–21 seasons. Keeping them would have reduced the dataset from 180 to 90 observations. Because Net Rating already accounts for pace differences across teams, this exclusion does not substantially reduce the predictive value of the model.

2.3 When

The dataset spans six NBA regular seasons from 2019–20 through 2024–25. All seasons used the standard 30-team league structure, though two seasons were affected by COVID-19 as described in the Limitations section below.

2.4 Where

All data was downloaded directly from Basketball Reference season pages. The team miscellaneous statistics tables for each season were exported as CSV files and read into R for cleaning and analysis.

2.5 Why

Basketball Reference was chosen because it is the most complete and consistently formatted free source of NBA team statistics available. It provides both traditional and advanced metrics in a uniform structure across seasons, which made it straightforward to combine multiple years of data without reconciling different column definitions or formats. No other public

source provides the same combination of variables at the team-season level without requiring a subscription.

2.6 How

Six season-level CSV files were downloaded, one per season, from Basketball Reference (Basketball Reference, 2020, 2021, 2022, 2023, 2024, 2025). Each file was read into R and filtered to remove the league average row that Basketball Reference appends at the bottom of each table. A `Playoff` binary indicator was created by detecting the asterisk that Basketball Reference appends to the names of teams that qualified for the postseason. Team names were then stripped of asterisks and trimmed, and all six files were combined using `bind_rows()`. The full cleaning code is included in the Code Appendix.

2.7 Limitations

Two seasons present structural differences from a standard NBA year. The **2019–20 season** was suspended in March 2020 due to COVID-19 and resumed in a neutral-site bubble in Orlando, Florida, with most teams playing between 63 and 75 games rather than a full 82. The **2020–21 season** was shortened to 72 games per team as a result of the delayed start caused by the pandemic. Win totals in both seasons are therefore lower than in a typical year, which may affect the models slightly since wins is one of the predictors. Additionally, teams within the same season are not fully independent because playoff qualification depends on conference standings — one team’s wins directly affect how many games another team loses. The dataset also covers only six seasons, so findings reflect the modern style of NBA play rather than broader historical trends.

3 Exploratory Data Analysis

Before fitting any models, we examined the structure and distributions of the dataset to understand what patterns might already be visible and whether classification would be feasible.

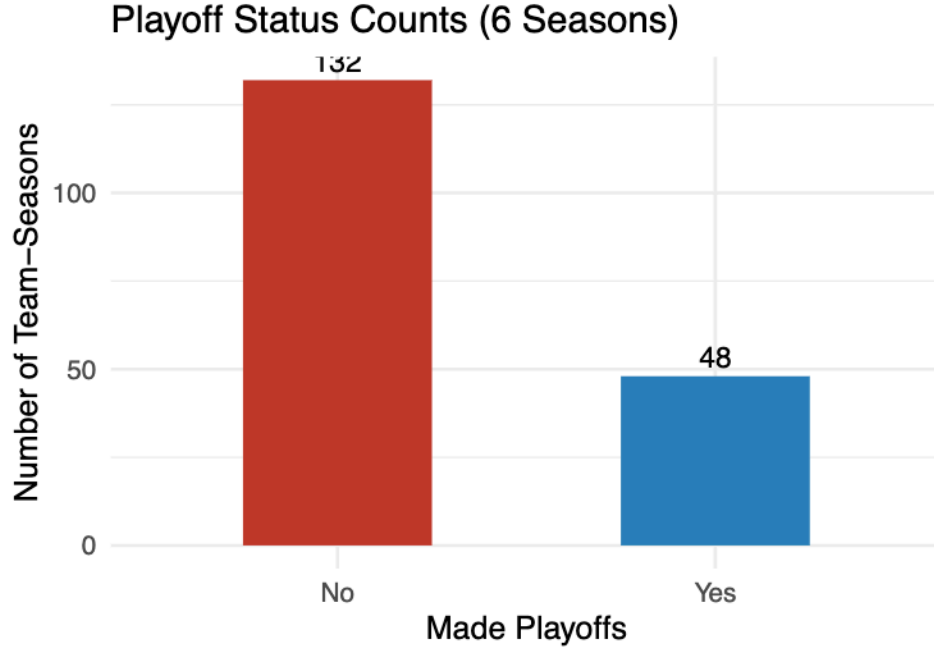


Figure 1: Playoff and non-playoff team counts across all six seasons. Non-playoff outcomes outnumber playoff outcomes roughly 2.75 to 1, reflecting the NBA’s structure of 16 qualifying teams out of 30.

Across six seasons, 48 team-seasons resulted in a playoff appearance and 132 did not. The NBA qualifies 16 of 30 teams each season — eight per conference — so non-playoff outcomes outnumber playoff outcomes by a ratio of roughly 2.75 to 1. This imbalance is moderate and is why we report sensitivity and specificity in addition to overall accuracy, since accuracy alone can be misleading when outcomes are not equally common.

Table 1: Playoff and Non-Playoff Counts by Season

Season	Did Not Make Playoffs	Made Playoffs
2019-20	30	NA
2020-21	30	NA
2021-22	30	NA
2022-23	14	16
2023-24	14	16
2024-25	14	16

Each season contributes exactly 30 observations, confirming the dataset is consistently structured across all six years. The season breakdown confirms no data is missing and that the

playoff indicator was applied correctly.

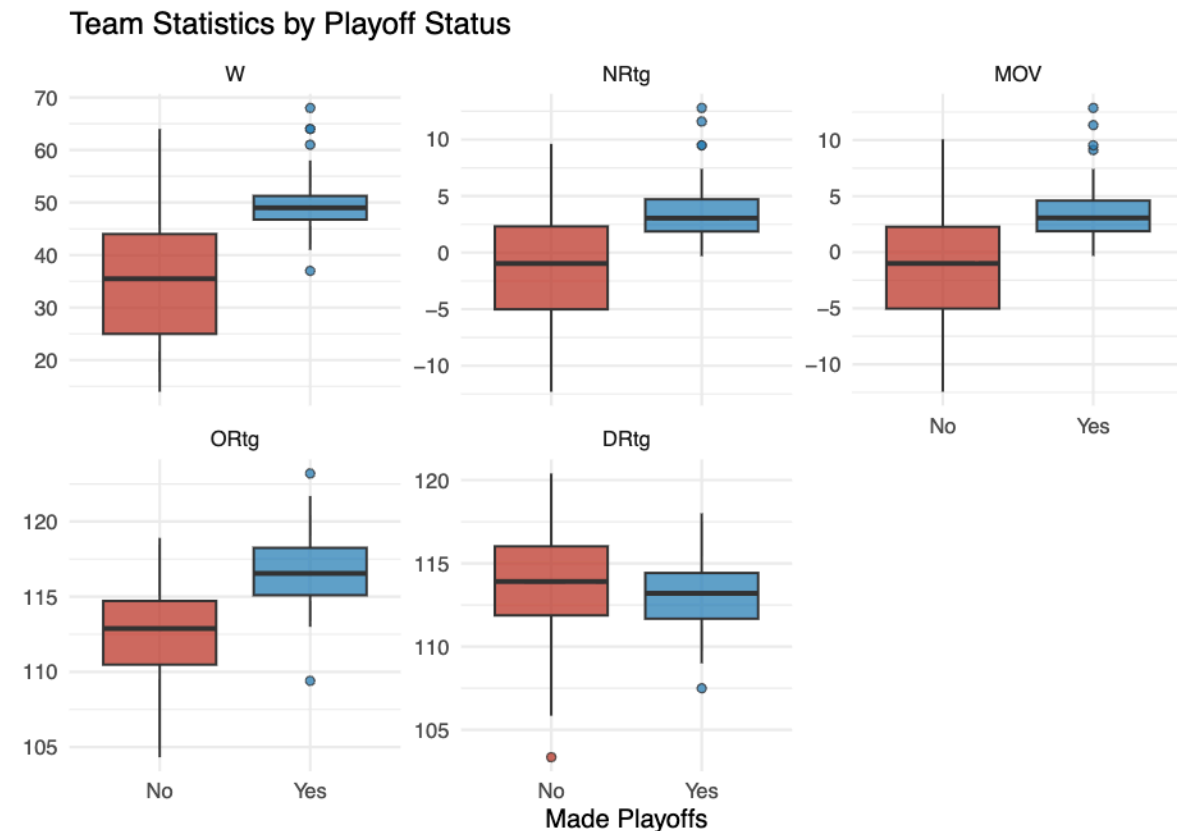


Figure 2: Distribution of five key team statistics by playoff status. Playoff teams show clear and consistent separation from non-playoff teams on efficiency and win-based metrics. For Defensive Rating, lower values indicate better defensive performance.

The boxplots show that playoff teams are clearly separated from non-playoff teams on wins, net rating, and margin of victory. Offensive rating also shows meaningful separation. Defensive rating is interpreted differently — lower values indicate a stronger defense — and playoff teams tend to cluster at lower defensive ratings, though the distributions overlap more than for net rating or wins. Overall, the separation across all five metrics suggests that multiple predictors carry useful signal for classification.

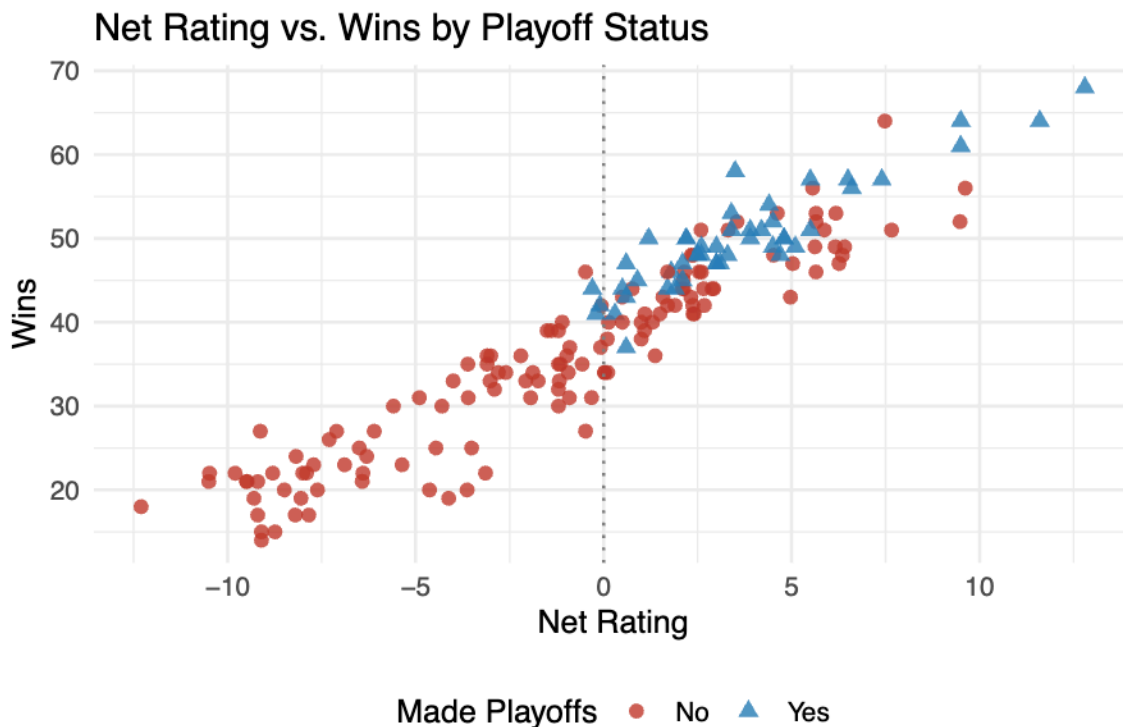


Figure 3: Net rating versus wins, colored by playoff status. Nearly every team with a positive net rating made the playoffs, confirming net rating as one of the strongest individual predictors.

The scatterplot confirms a strong near-linear relationship between net rating and wins. Nearly all teams with a positive net rating made the playoffs. This makes intuitive sense: teams that consistently outscore opponents on a per-possession basis tend to win enough games to qualify. The few exceptions are likely explained by conference competitiveness or the shortened COVID-affected seasons. This scatterplot also highlights an important modeling challenge: because net rating, wins, and margin of victory all measure similar things, they are highly correlated — a complication we address in the discussion of logistic regression below.

4 Machine Learning Methods

We used three supervised classification algorithms to predict playoff qualification. All models share the same setup: 70% of team-season observations were used for training and the remaining 30% for testing. Within the training set, we applied 5-fold cross-validation to tune each model and reduce overfitting. The response variable `Playoff` is binary (Yes or No), making this a classification rather than a regression problem. We evaluate each model using accuracy,

sensitivity, specificity, and AUC (area under the ROC curve), because together these metrics give a complete picture of model performance when classes are not equally balanced.

4.1 Logistic Regression

What it is and how it works: Logistic regression is the regression-based algorithm used in this analysis. Rather than predicting a number, it estimates the *probability* that a team makes the playoffs given its statistics. It works by combining predictors with learned weights, summing them, and then transforming the result through the logistic function — an S-shaped curve — to produce a value between 0 and 1. If the predicted probability exceeds 0.5, the team is classified as a playoff team; otherwise it is not.

How to interpret results: A positive coefficient on a predictor means that higher values of that variable are associated with a greater probability of making the playoffs. A negative coefficient means the opposite. For example, a positive coefficient on net rating would confirm that teams with better net ratings are more likely to qualify, which aligns with what the EDA already showed.

Strengths: Logistic regression is one of the most interpretable classification methods. Its coefficients directly communicate the direction and relative magnitude of each predictor’s relationship with playoff probability. It also outputs calibrated probabilities, which are useful for ranking teams rather than just labeling them.

Weaknesses: The main limitation here is multicollinearity. Wins, margin of victory, and net rating are all highly correlated because they measure similar aspects of team quality. When predictors are correlated, logistic regression can produce unstable and hard-to-interpret coefficient estimates. Logistic regression also assumes a linear relationship between predictors and the log-odds of the outcome, which may not hold perfectly.

When it is most appropriate: Logistic regression is the best choice when interpretability is the top priority, when predictors are not strongly correlated with each other, and when the analyst needs probability estimates rather than just class labels.

Table 2: Logistic Regression Confusion Matrix (Test Set)

	No	Yes
No	31	0
Yes	8	14

The logistic regression model correctly classified **84.9%** of all teams in the test set. Of the actual playoff teams in the test set, it correctly identified **100%** (sensitivity). Of the actual non-playoff teams, it correctly identified **79.5%** (specificity). The confusion matrix shows the exact breakdown of correct and incorrect predictions in each category.

4.2 Decision Tree (CART)

What it is and how it works: A decision tree classifies teams by learning a set of yes/no rules from the training data. The algorithm starts with all teams grouped together and asks: which single predictor and threshold best separates playoff from non-playoff teams? It makes that split, then repeats the process in each resulting group, continuing until it reaches a stopping point. The result is a flowchart-like structure where each team follows a path from the top of the tree to a leaf node that assigns a predicted class. This method was developed by Breiman et al. (1984) and the implementation used here follows the CART framework.

How to interpret results: You can follow the tree from top to bottom. At each internal node, the tree asks a question about a statistic — for example, “Is the team’s net rating above 2.5?” Teams that answer yes go one direction; teams that answer no go the other. Eventually every team arrives at a leaf that says either “Playoff: Yes” or “Playoff: No.”

Strengths: Decision trees are the most interpretable of the three methods. No statistics background is required to follow the logic. They also handle correlated predictors naturally — rather than estimating all predictors simultaneously, they simply pick the most useful one at each split.

Weaknesses: Decision trees are unstable. A small change in the training data can lead to a noticeably different tree structure. They can also overfit when grown too deep, which is why we used cross-validation to tune the complexity parameter and find the right tree size.

When it is most appropriate: A decision tree is the best choice when the audience needs to understand and follow the logic of individual predictions, when a simple rule-based output is more useful than a probability, or when there is a need to explain the model to a non-technical audience.

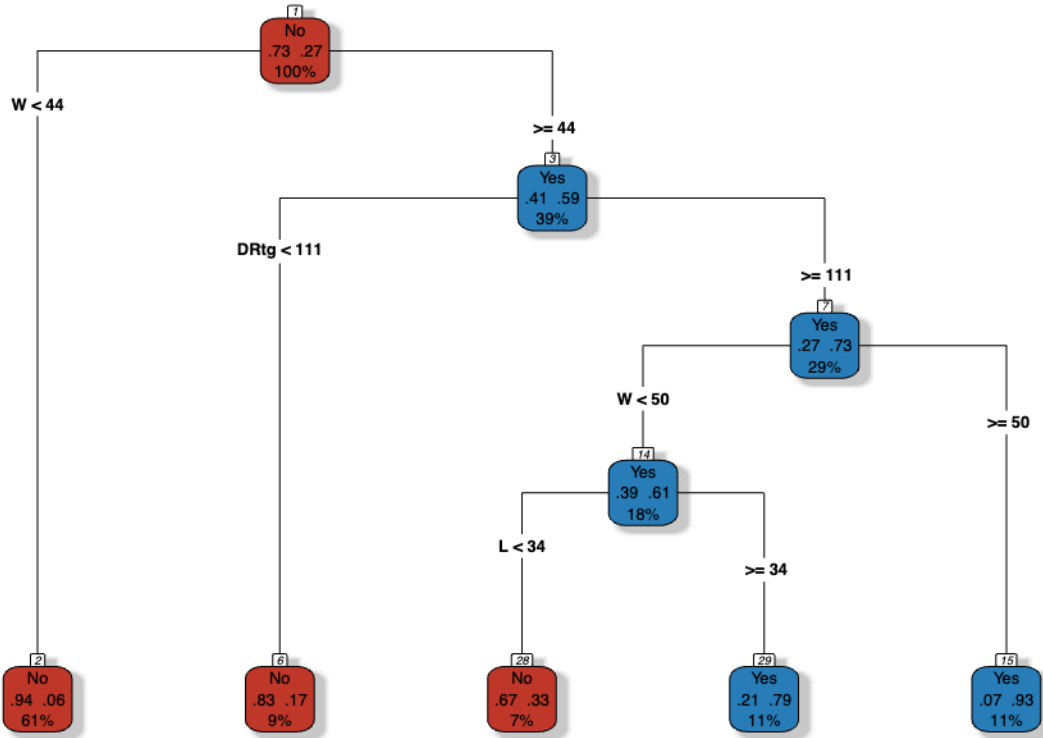


Figure 4: Pruned decision tree for playoff prediction. Each node shows the predicted class, the probability of being a playoff team, and the percentage of observations that reach that node.

Table 3: Decision Tree Confusion Matrix (Test Set)

	No	Yes
No	32	3
Yes	7	11

The decision tree correctly classified **81.1%** of teams in the test set, with a sensitivity of **78.6%** and a specificity of **82.1%**. The tree diagram shows which statistics the algorithm relied on most — predictors near the top of the tree, at the root node, were found to be the most useful for separating playoff from non-playoff teams.

4.3 Random Forest

What it is and how it works: A random forest is an ensemble method that extends the decision tree idea. Instead of building one tree, it builds 500 trees, each trained on a different random bootstrap sample of the data and using a random subset of predictors at each split (Breiman, 2001). This intentional randomness prevents all trees from being identical and reduces the instability that plagues individual trees. To classify a team, every tree casts a vote on the outcome, and the majority vote determines the final prediction.

How to interpret results: Individual trees in a random forest cannot be easily inspected the way a single decision tree can. However, the model produces *variable importance* scores that measure how much each predictor contributed to accurate predictions across all 500 trees. Higher scores indicate statistics that were more consistently useful for separating playoff from non-playoff teams — these scores are the most direct answer to our research question.

Strengths: Random forests typically produce the most accurate and stable predictions of the three methods. By aggregating many trees, they cancel out the errors that any individual tree might make. They handle correlated predictors and nonlinear relationships well, and their variable importance output is directly interpretable for our research question.

Weaknesses: A random forest is much harder to explain than a single decision tree. There is no simple flowchart to show a general audience, and individual predictions cannot be easily traced through the model. Training 500 trees also takes more computational time than logistic regression or a single tree.

When it is most appropriate: A random forest is the best choice when predictive accuracy is the primary goal, when the relationships between predictors and the outcome may be complex or nonlinear, and when the analyst is willing to treat the model as a “black box” and rely on summary outputs like variable importance scores rather than individual prediction pathways.

Table 4: Random Forest Confusion Matrix (Test Set)

	No	Yes
No	35	3
Yes	4	11

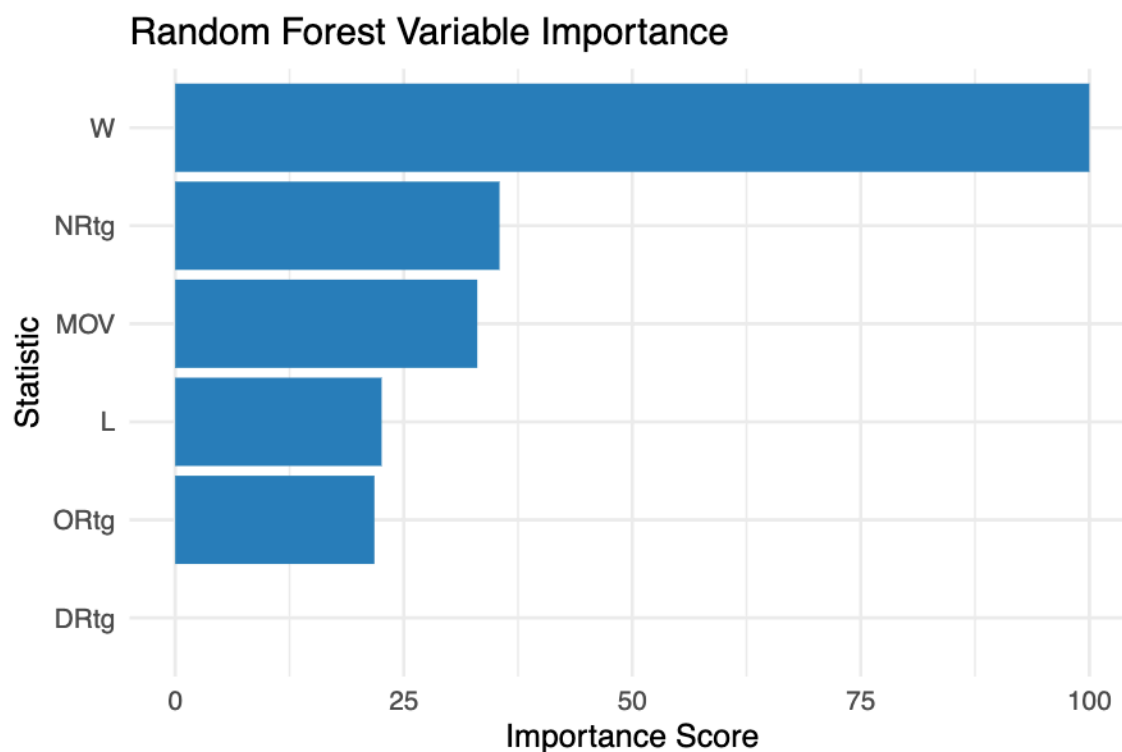


Figure 5: Random forest variable importance. The importance score reflects how much each statistic contributed to accurate predictions across all 500 trees. Higher values indicate greater predictive contribution to correctly classifying playoff teams.

The random forest correctly classified **86.8%** of teams in the test set, with a sensitivity of **78.6%** and a specificity of **89.7%**. The variable importance plot provides the most direct answer to the research question by showing which statistics most consistently helped the model distinguish playoff teams from non-playoff teams across all 500 trees.

5 Model Comparison

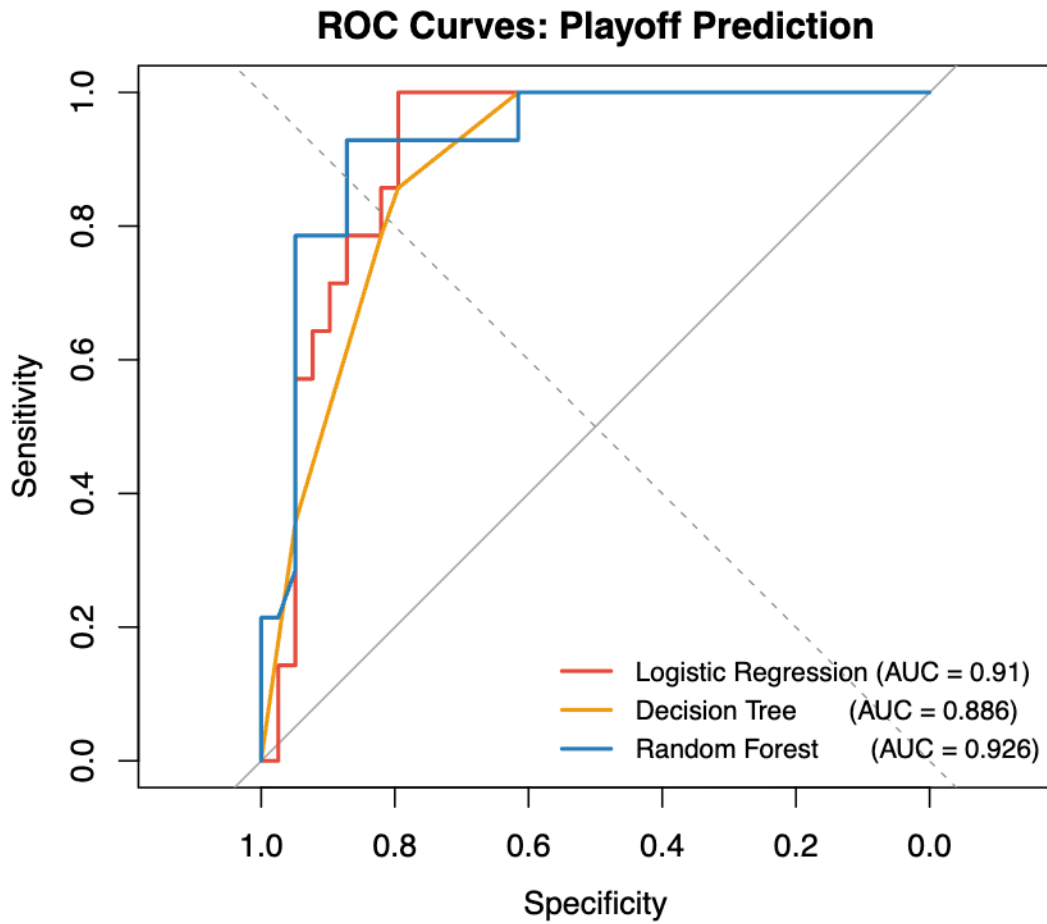


Figure 6: ROC curves for all three models. A curve that bends more sharply toward the top-left corner indicates better performance. The diagonal dashed line represents random guessing ($AUC = 0.5$). All three models substantially outperform random chance.

Table 5: Model Performance Summary on the Test Set

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.849	1.000	0.795	0.910
Decision Tree	0.811	0.786	0.821	0.886
Random Forest	0.868	0.786	0.897	0.926

The strongest overall model by AUC was **Random Forest**. Accuracy measures the share of all predictions that were correct, but it can overstate performance when classes are imbalanced — a model that always predicted “No Playoffs” would achieve 73% accuracy without learning anything useful. Sensitivity captures how well a model identifies actual playoff teams, specificity measures how well it identifies non-playoff teams, and AUC summarizes discriminative ability across all possible classification thresholds. Together, these four metrics give a complete picture of each model’s performance.

Comparing the three approaches: Each model involves a tradeoff between interpretability and predictive power. Logistic regression (Kuhn, 2008) is the most interpretable — its coefficients communicate the direction and magnitude of each predictor’s relationship with playoff probability — but it is the most sensitive to multicollinearity among predictors. The decision tree is the easiest to explain to a general audience because it produces a visible flowchart, but individual trees are unstable and can change substantially with small shifts in training data. The random forest (Breiman, 2001) sacrifices interpretability for accuracy and stability — by averaging 500 trees, it reduces the variance that makes individual trees unreliable and produces the most trustworthy variable importance rankings.

6 Discussion

6.1 What We Learned

The most important finding from this project is that NBA playoff qualification is strongly and consistently predictable from a small set of regular season team statistics. Across all three models, the highest-importance predictors were **net rating**, **wins**, and **margin of victory**. These metrics are closely related — they all measure how much a team outperforms its opponents over a full season — but they capture that information at different levels of refinement. Wins are the most direct measure but are sensitive to schedule strength and conference competitiveness. Margin of victory adjusts for context on a game-by-game basis. Net rating goes furthest by accounting for the number of possessions played, making it the most reliable efficiency summary available from team-level data.

The random forest variable importance plot provides the clearest and most direct answer to the research question. Statistics that summarize overall team dominance rank highest regardless of whether that dominance comes from offense or defense. This suggests that what separates playoff teams is not specialization on one side of the ball but sustained superiority across a full 70-plus game season.

The EDA supported this conclusion before any model was fit. The scatterplot of net rating versus wins showed near-perfect separation between playoff and non-playoff teams along the net rating axis. A data analyst without any machine learning tools could examine a team’s net rating in January and already have a strong prediction of whether they will make the playoffs — the machine learning models in this project quantify and formalize that intuition.

6.2 Comparing the Methods and Decisions Made

We chose logistic regression as our regression-based method because it is the standard approach for binary classification and is directly relevant to the course material. We chose the decision tree (CART) and random forest as our non-regression methods because they provide a useful contrast: the decision tree maximizes interpretability while the random forest maximizes stability and accuracy.

The decision to exclude Pace and Free Throw Rate was driven entirely by data availability. Including them would have reduced the dataset from 180 to 90 observations by removing the two COVID-affected seasons. Since Net Rating already incorporates pace differences across teams, the loss of predictive information was judged to be minimal relative to the benefit of keeping a larger and more representative dataset.

We used 5-fold cross-validation during training rather than relying on the single 70/30 split alone, because cross-validation provides a more reliable estimate of how each model generalizes to new data by averaging results across multiple held-out subsets of the training data.

6.3 Recommendations for Future Work

Several extensions would strengthen this analysis. First, expanding the dataset to cover ten or more seasons would provide more training examples and reduce the influence of unusual years like the COVID bubble seasons. Second, adding player-level data — such as a measure of roster quality, star player availability, or games missed to injury — could substantially improve predictions, because team statistics alone do not capture roster composition. Third, applying principal component analysis (PCA) before logistic regression could address the multicollinearity issue by transforming the correlated predictors into independent components, which would make coefficient estimates more stable and interpretable.

7 Conclusion

This project found that NBA playoff qualification is highly predictable from regular season team statistics. The most important predictors across all three models were net rating, wins, and margin of victory — all of which capture overall team dominance rather than any single dimension of play. The random forest achieved the strongest predictive performance, while the decision tree provided the most interpretable structure for a general audience, and logistic regression served as a competitive and interpretable baseline. The results support a clear conclusion: playoff teams are not defined by a particular style of play, but by their ability to consistently outperform opponents across a full regular season on both ends of the floor.

8 Author Contributions

- **Varun Gullanki:** Data collection, file merging, and cleaning script; data provenance section
- **Amal Parekh:** Exploratory data analysis and all visualizations
- **Luke Harrington:** Model implementation, cross-validation setup, and results interpretation
- **Marlon Dubreuil:** Discussion, conclusion, report editing, and unified writing voice

(Update with actual contributions before submitting.)

9 AI Use Statement

Generative AI (Claude, Anthropic) was used as a reference tool to help structure the Quarto document, debug R rendering errors, and improve wording in several sections of the report. All AI-assisted content was reviewed, verified, and edited by the group. The analysis decisions, interpretations, variable choices, and conclusions are the group's own work. All AI use is documented here in accordance with the course policy.

10 References

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11 Code Appendix

All code is included below for reproducibility. No code or raw output appears in the body of the report. The data cleaning script (run separately before rendering) is shown first, followed by the full analysis code used in this document.

```
# =====
# PART 1: DATA CLEANING SCRIPT
# Run this script separately to produce nba_all_seasons_clean_final.csv
# before rendering the QMD report.
# =====

library(tidyverse)

# Read each season file (skip = 1 removes Basketball Reference's extra title row)
df_19_20 <- read_csv("19_20_data.csv", skip = 1, show_col_types = FALSE)
df_20_21 <- read_csv("20_21_data.csv", skip = 1, show_col_types = FALSE)
df_21_22 <- read_csv("21_22_data.csv", skip = 1, show_col_types = FALSE)
df_22_23 <- read_csv("22_23_data.csv", skip = 1, show_col_types = FALSE)
df_23_24 <- read_csv("23_24_data.csv", skip = 1, show_col_types = FALSE)
df_24_25 <- read_csv("24_25_data.csv", skip = 1, show_col_types = FALSE)

# Function to clean one season:
# - Removes the league average summary row
# - Creates Playoff indicator: 1 if team name has asterisk, 0 otherwise
# - Strips asterisk and leading/trailing whitespace from team names
# - Adds a season label column for later identification
clean_season <- function(df, season_label) {
  df %>%
    filter(Team != "League Average") %>%
    mutate(
      Playoff = ifelse(grepl("\\\\*", Team), 1, 0), # detect asterisk BEFORE removing it
    )
}
```

```

    Team    = gsub("\\\\*", "", Team),
    Team    = str_trim(Team),
    Season  = season_label
  )
}

# Apply cleaning function to each season
df_19_20 <- clean_season(df_19_20, "2019-20")
df_20_21 <- clean_season(df_20_21, "2020-21")
df_21_22 <- clean_season(df_21_22, "2021-22")
df_22_23 <- clean_season(df_22_23, "2022-23")
df_23_24 <- clean_season(df_23_24, "2023-24")
df_24_25 <- clean_season(df_24_25, "2024-25")

# Stack all six seasons into one dataset (180 rows)
nba_all <- bind_rows(df_19_20, df_20_21, df_21_22, df_22_23, df_23_24, df_24_25)

# Select only the columns used in modeling
# Pace and FTr excluded because they are missing for 2019-20 and 2020-21
# Net Rating already captures pace-adjusted efficiency, so this exclusion
# does not substantially reduce predictive power
nba_all_clean <- nba_all %>%
  select(Team, Season, Playoff, W, L, MOV, ORtg, DRtg, NRtg)

# Save final cleaned dataset
write_csv(nba_all_clean, "nba_all_seasons_clean_final.csv")

# =====
# PART 2: ANALYSIS CODE
# This is the code that runs inside the QMD report.
# =====

library(tidyverse)
library(caret)
library(rpart)
library(rpart.plot)
library(randomForest)
library(knitr)
library(pROC)

set.seed(123)

```

```

# Load pre-cleaned dataset and apply type conversions
nba_clean <- read_csv("nba_all_seasons_clean_final.csv", show_col_types = FALSE) %>%
  mutate(
    Playoff = factor(ifelse(Playoff == 1, "Yes", "No"), levels = c("No", "Yes")),
    Season = as.factor(Season),
    Team = as.character(Team),
    across(c(W, L, MOV, ORtg, DRtg, NRtg), as.numeric)
  )

# Drop Team and Season (identifiers, not predictors); drop_na as safety check
nba_model <- nba_clean %>%
  select(Playoff, W, L, MOV, ORtg, DRtg, NRtg) %>%
  drop_na()

# 70/30 train/test split, stratified on Playoff to preserve class ratio
train_index <- createDataPartition(nba_model$Playoff, p = 0.70, list = FALSE)
train_data <- nba_model[train_index, ]
test_data <- nba_model[-train_index, ]

# 5-fold cross-validation; ROC used as the primary tuning metric
ctrl <- trainControl(
  method = "cv",
  number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary
)

# -- Logistic Regression -----
# Regression-based classifier; estimates playoff probability via logistic fn
log_model <- train(
  Playoff ~ ., data = train_data,
  method = "glm", family = "binomial",
  trControl = ctrl, metric = "ROC"
)
log_pred <- predict(log_model, test_data)
log_probs <- predict(log_model, test_data, type = "prob")[, "Yes"]
log_cm <- confusionMatrix(log_pred, test_data$Playoff, positive = "Yes")

# -- Decision Tree (CART) -----
# Non-regression classifier; learns if-then rules from training data
# tuneLength = 10 tries 10 values of the complexity parameter cp
tree_model <- train(

```

```

Playoff ~ ., data = train_data,
method = "rpart", trControl = ctrl,
metric = "ROC", tuneLength = 10
)
tree_pred <- predict(tree_model, test_data)
tree_probs <- predict(tree_model, test_data, type = "prob")[, "Yes"]
tree_cm <- confusionMatrix(tree_pred, test_data$Playoff, positive = "Yes")

# -- Random Forest -----
# Non-regression ensemble: builds 500 trees, each on a random bootstrap sample
# importance = TRUE enables variable importance calculation after training
rf_model <- train(
  Playoff ~ ., data = train_data,
  method = "rf", trControl = ctrl,
  metric = "ROC", importance = TRUE, ntree = 500
)
rf_pred <- predict(rf_model, test_data)
rf_probs <- predict(rf_model, test_data, type = "prob")[, "Yes"]
rf_cm <- confusionMatrix(rf_pred, test_data$Playoff, positive = "Yes")

# -- ROC curves -----
roc_log <- roc(test_data$Playoff, log_probs,
              levels = c("No", "Yes"), direction = "<", quiet = TRUE)
roc_tree <- roc(test_data$Playoff, tree_probs,
              levels = c("No", "Yes"), direction = "<", quiet = TRUE)
roc_rf <- roc(test_data$Playoff, rf_probs,
              levels = c("No", "Yes"), direction = "<", quiet = TRUE)

# -- Summary performance table -----
results_df <- tibble(
  Model = c("Logistic Regression", "Decision Tree", "Random Forest"),
  Accuracy = round(c(log_cm$overall["Accuracy"],
                    tree_cm$overall["Accuracy"],
                    rf_cm$overall["Accuracy"]), 3),
  Sensitivity = round(c(log_cm$byClass["Sensitivity"],
                      tree_cm$byClass["Sensitivity"],
                      rf_cm$byClass["Sensitivity"]), 3),
  Specificity = round(c(log_cm$byClass["Specificity"],
                      tree_cm$byClass["Specificity"],
                      rf_cm$byClass["Specificity"]), 3),
  AUC = round(c(auc(roc_log), auc(roc_tree), auc(roc_rf)), 3)
)

```

```

# -- Variable importance -----
# rowMeans used because caret's varImp output column name varies by R/caret version
# This ensures the code is robust regardless of whether caret returns "Overall"
# or class-specific columns (e.g., "Yes", "No")
rf_importance <- varImp(rf_model)$importance %>%
  as.data.frame() %>%
  rownames_to_column("Statistic") %>%
  mutate(Overall = rowMeans(across(where(is.numeric)), na.rm = TRUE)) %>%
  arrange(desc(Overall))

```